Linux Sample Questions Solutions

1	The **pwd** command stands for **"Print Working Directory."** It is used to display the full path name of the current directory the user is in. ls lists files, cd changes directory, and man displays the manual page for a command.
2	**Linux** (specifically the kernel) was developed by Linus Torvalds as a free, open-source re-implementation, or clone, of the **UNIX** principles and API (POSIX standard). It is not a direct descendant of the original AT&T UNIX source code, unlike true UNIX variants (like Solaris or AIX).
3	In Linux file permissions, the octal value is calculated by adding the values: Read (4), Write (2), and Execute (1). **600** translates to **rwx** for the user (owner) and **---** for the group and others. r (4) + w (2) = 6. The three digits correspond to **User (owner)**, **Group**, and **Others**.
4	The first line of a shell script is called the **shebang**, which tells the operating system which interpreter to use to execute the script. For a Bash script, this is almost universally #!/bin/bash, pointing to the location of the Bash interpreter executable.
5	**APT** (Advanced Package Tool) is the core package management system for Debian and its derivatives (e.g., Ubuntu, Mint). RPM, DNF, and YUM are primarily used on Red Hat-based distributions (e.g., Fedora, CentOS, RHEL).
6	A **process** is the operating system's abstraction of a running program. It includes the program's code, data, stack, and its associated resources. A **daemon** is a specific type of background process, and the **kernel** and **shell** are components of the operating system itself.
7	**Paging** is a memory management scheme that avoids the problem of external fragmentation by dividing the logical memory into fixed-size blocks (**pages**) and physical memory into same-sized blocks (**frames**), allowing a process to be scattered throughout physical memory.
8	The **Shortest Job First (SJF)** scheduling algorithm associates with each process the length of its next CPU burst and uses these lengths to schedule the process with the shortest time. In its non-preemptive form, a process runs to completion once it starts.
9	In traditional programming (e.g., C/C++), a do-while loop executes at least once. However, standard Bash shell scripting only provides **for**, **while**, and **until** loops, all of which check the conditional expression *before* the first iteration. Therefore, if the condition is initially false, they will not execute even once.
10	**Ext4** (Fourth Extended Filesystem) is the current standard and a significant improvement over Ext3, introducing features like extents, multi-block allocation, and delayed allocation, leading to better performance and scalability.
11	The correct syntax for numerical comparison within the [[...]] test construct in Bash is to use the operator **-ge** (greater than or equal to). The full structure requires then to introduce the block of commands. [...] or [[...]] is the correct test structure, and -ge is the correct operator.

12	**Arch Linux** is known for its rolling-release model, minimal base system, and unique package manager, **pacman**. Fedora and CentOS are RPM-based, and Debian is a fixed-release distribution that uses the APT package manager.
13	The **Waiting (or Blocked)** state is where a process resides when it is executing code but is waiting for some event to occur, such as the completion of an I/O request, the receipt of a signal, or the availability of a resource.
14	**Virtual Memory** allows processes to be larger than physical memory by mapping a logical address space to a potentially smaller physical address space. It achieves this by bringing only the required pages/segments of a process into RAM, swapping out others to secondary storage (disk).
15	A **critical section** is a segment of code where a process or thread accesses a **shared resource** (like a variable or file). To maintain data integrity and prevent race conditions, only one process/thread must be allowed to execute its critical section at any given time (mutual exclusion).
16	The **rpm** command is the low-level tool for Red Hat Package Manager. The options **-q** (query) and **-a** (all packages) combine to list every package installed via RPM. dpkg -l and apt list --installed are used for APT/Debian systems.
17	The chmod command is used to change file permissions. In symbolic mode, **o** stands for **others**, **+** means **add** permission, and **w** stands for **write**. The command grants write permission to users who are neither the owner nor in the file's group.
18	The **for** loop in Bash is typically used to iterate over a list of words or items provided in a sequence (e.g., for i in item1 item2 item3; do ... done). while and until loops execute as long as a condition is true or until a condition becomes true, respectively.
19	The main issue with implementing SJF is that it requires knowing the **length of the next CPU burst** in advance, which is generally impossible to predict accurately in a real system. Estimations are used in practice, but the theoretical requirement is the primary challenge.
20	The **Process Control Block (PCB)** is a data structure containing information about the process's current state (registers, program counter, scheduling priority, memory management info). The complete **executable code** of the program itself is stored in memory (text segment), not within the PCB data structure.
21	In Linux, file descriptors 0, 1, and 2 correspond to standard input (stdin), standard output (stdout), and standard error (stderr), respectively. The operator **2>** specifically redirects the output of file descriptor 2 (**stderr**) to the specified file.
22	A **Semaphore** is a classic synchronization tool, which is essentially an integer variable. It is used to solve the critical-section problem by using two indivisible (atomic) operations: **wait()** (or P) and **signal()** (or V) to control access to resources.
23	A **Page Fault** is an interrupt that occurs when a process attempts to access a

	page that is mapped in the address space but is not currently loaded into physical memory. The operating system must then load the required page from disk.
24	**XFS** (originally developed by SGI) is a journaling file system known for its scalability and high performance, especially with very large files and file systems. While it uses "extents" for allocation, it is typically just referred to as XFS.
25	The **until** loop in Bash is the inverse of the while loop. It executes the body of the loop **until** the condition becomes true. Therefore, the loop continues to run as long as the conditional test evaluates to **False** (a non-zero exit status).
26	The **Linux** kernel is a prominent example of collaborative and publicly visible development under the **open-source** model (specifically the GPL license). Traditional UNIX kernels were proprietary, closed-source, and vendor-specific.
27	**DNF** (Dandified YUM) is the next-generation package manager for RPM-based distributions like Fedora, RHEL 8+, and CentOS 8-Stream, having largely replaced the older **YUM** utility.
28	A **daemon** is a process that runs in the background, typically started at boot time, waiting for an event or condition to occur (e.g., a web server daemon waiting for a connection). They are often system services and have no controlling terminal.
29	In **Segmentation**, the memory space is divided into logical segments of *variable* size. Because the free memory is split into varying-sized holes, external fragmentation can occur, where small free spaces cannot be combined to satisfy a request for a large segment. Paging avoids this by using fixed-size blocks.
30	In Round Robin, if a process completes its burst before the time quantum Q is exhausted, it releases the CPU, and the CPU scheduler immediately performs a **context switch** to select the next process from the ready queue.
31	A deadlock is characterized by four necessary and jointly sufficient conditions: **Mutual Exclusion**, **Hold and Wait**, **No Preemption**, and **Circular Wait**. The *absence* of Preemption (**No Preemption**) is a condition for deadlock; therefore, **Preemption** itself is not a necessary condition.
32	A context switch requires saving the state of the old process and loading the state of the new process. When switching to a different address space (a different process), the memory management hardware's **Translation Lookaside Buffer (TLB)**, which caches page table entries, often needs to be

	flushed or invalidated to prevent the new process from using stale address translations, a costly operation.
33	The **shift** command in Bash is used to shift positional parameters ($1, $2, etc.) to the left. shift 2 means the first two arguments ($1 and $2) are discarded, and the parameter that was previously $3 now becomes the new $1. Thus, arg3 becomes the new $1.
34	The number of pages in the logical address space is determined by the formula: $2^{(\text{Logical Address Bits} - \text{Page Offset Bits})}$. ● Logical Address size = 32 bits ● Page size = $4\text{KB} = 2^{12}\text{ bytes}$. ● Page Offset Bits = 12 bits. ● Number of Page Entries = $2^{(32 - 12)} = 2^{20}$. Since each page requires one entry in the page table, the table has 2^{20} entries.
35	A **mount point** is an empty directory in the currently accessible file system (the host file system) where the root of another file system (from a disk partition or device) is attached, making the contents of the new file system accessible to the user via that directory path.
36	A **Mutex (Mutual Exclusion)** is a locking mechanism, often binary (locked/unlocked), used to ensure only one thread/process can access a critical section. A **Counting Semaphore** is an integer-value object used to control access to a pool of multiple identical resources; its value represents the number of available resources.
37	The **Convoy Effect** occurs primarily in **FCFS** (First-Come, First-Served) scheduling. If a single, CPU-bound process with a very long CPU burst arrives first, all subsequent processes (even very short ones) must wait for it to complete, leading to poor CPU utilization for I/O-bound processes.
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39	Front-end tools like dnf or apt automatically check for and resolve **dependencies** (other packages required by the main one). Manual installation using a low-level command like rpm -i or dpkg -i generally skips this

	dependency resolution, requiring the user to manually install any missing packages first, which is the primary reason it's discouraged.
40	In a **monolithic kernel** architecture, the entire operating system, including the CPU scheduler, memory manager, file systems, and all device drivers, operates in the privileged **kernel space** as one large, single program. While Linux has a degree of modularity (loadable kernel modules), its fundamental structure is monolithic, unlike microkernels (where services run in user space).

Topics

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• Introduction to the Linux Operating System
• History of Linux.
• Linux vs. Unix
• CentOS8-Stream Installation
• Basic Commands.
Introduction to Shell and Shell Scripting
• Basic Shell Commands
• Writing Shell Scripts
• Conditional statements.
• Loops: for, while, until
Create a basic shell script.
• Basic Shell Commands
• Write scripts using if, if-else, and if-elif-else.
• Loops
• Process
• Types of Processes
• Process States

• Process Control Block (PCB)
• Context Switching
Scheduling & synchronization
• CPU Scheduling
• Scheduling Algorithms
• Synchronization Basics
• Synchronization Techniques
Memory management
• Memory Allocation Techniques
• Paging
• Segmentation
• Virtual Memory
File Systems
• File System Types
• Linux File Systems (Ext2/3/4,XFS).
• File and Directory Permissions
• Explore and list the file systems used in your system.

• Format a partition using Ext4 and mount it on a directory.
• Change file permissions using chmod.
Package Managers (RPM, APT, etc.).
• Types of Package Managers

• RPM (Red Hat-
based systems)

• APT(Debian-
based systems)

• Differences
between RPM

and APT,
YUM, DNF
basics
Install and remove a package using both RPM and APT.
• Install a .rpm file manually and verify the installation.
• Create a shell script to automate the installation of a software suite using a package manager.